Problem 1. Completely read the Homework Guidelines that are posted on Canvas.

Solution. As I said, I like hiking. □

Problem 2. Let x be an even integer and let n be a natural number. Prove that x^n is an even integer.

Proof. Since x is an even integer, we know that there exists some integer k such that $x = 2k$. Now we compute

$$x^n = (2k)^n = 2(2^{n-1}k^n).$$

Since n is a natural number, $2^{n-1}k^n$ is an integer. Thus, the expression $x^n = 2(2^{n-1}k^n)$ shows that x^n is an even integer, as desired. □

Problem 3. Let $n \in \mathbb{N}$. (In other words, n is a natural number.)

1. Make at least two conjectures about factorizations/primeness of $3^n - 1$.
2. Make at least two conjectures about factorizations/primeness of $3^n - 2^n$.

For each of these, you can make a table like in the example we did on the first day of class to guide your conjectures. You do not need to prove your conjectures.

Note: A conjecture should be something you suspect is true based on a reasonable amount of evidence, but you don't know for sure. Technically, "$3^n - 1$ is prime for every value of n" might be a conjecture, but it is not a very good one, because $3^2 - 1$ is not prime, so the conjecture is clearly false. Also, technically "$3^n - 1$ is an integer" might be a conjecture, but not a good one since it is obviously true and does not address primeness.

Solution. After looking at some data, we can say the following conjectures (there may be others). Note that you should have shown your data, since this is part of showing your work. If you haven't shown your data, the reader does not know how you came up with your conjectures. A conjecture should be a clearly stated mathematical statement that is consistent with your data and that you suspect is true. If we don't have any data, we could conjecture things like "For all $n \in \mathbb{N}$, $3^n - 1$ is negative." This is a clear mathematical statement, but is not a conjecture since we have no reason to suspect that it is true, and in fact just looking at a one example quickly disproves it.

Conjecture A1. For all $n \in \mathbb{N}$, $3^n - 1$ is always divisible by 2.

Conjecture A2. If $n > 1$, then $3^n - 1$ is not prime.

(In fact, you could probably prove both of these conjectures! But since we haven't proved them yet, they are just conjectures for now.)

Conjecture B1. If a natural number n is not prime, then $3^n - 2^n$ is not prime.

Conjecture B2. If n is even, then $3^n - 2^n$ is divisible by 5. □

Problem 4. Let P be the statement "I went hiking last weekend", let Q be the statement "Greek olives are from Italy," and R be the statement "Birds are not real." Write the following logical expressions in full English sentences.

1. $\neg(P \vee Q) \wedge R$.
2. $(Q \vee \neg R) \vee (\neg Q \wedge P)$.
3. $\neg[P \wedge (\neg R \vee \neg Q)] \vee \neg Q$.

Solution.

1. Using De Morgan's law $\neg(P \vee Q) \equiv \neg P \wedge \neg Q$, this becomes "I did not go hiking last weekend, Greek olives are not from Italy, and birds are not real."

2. "Either Greek olives are from Italy or birds are real, or both Greek olives are not from Italy and I went hiking last weekend."

3. This expression simplifies to $\neg P \vee \neg Q \vee (R \wedge Q)$[1]. In English, this says "Either I did not go hiking last weekend, or Greek olives are not from Italy, or both Birds are not real and Greek olives are from Italy."

□

Problem 5. For the following sentences, assign P, Q, etc. to each statement and write these as logical expressions. (In other words, do the opposite of the previous problem.)

1. Either Kathy has been fired or she has quit her job, and she's moving to Argentina and she's never coming back.

2. I can't type very fast, but this document will be finished eventually.

3. Snoopy is either hungry and he is not tired, or he is not hungry and he is angry.

Solution.

1. Let P = "Kathy has been fired," Q = "Kathy has quit her job," R = "Kathy is moving to Argentina," and S = "Kathy is never coming back." Then the above expression is $(P \vee Q) \wedge R \wedge S$.

2. Let P = "I can't type very fast," and Q = "This document will be finished eventually." The expression is just $P \wedge Q$.

3. Let P = "Snoopy is hungry," Q = "Snoopy is tired," and R = "Snoopy is angry." Then the expression is $(P \wedge Q) \vee (\neg P \wedge R)$.

□

Problem 6. Make truth tables for the following formulas.

1. $\neg[P \wedge (Q \vee \neg P)]$.

2. $(P \wedge Q) \vee (\neg P \wedge \neg Q)$.

3. $(P \vee Q) \wedge (P \vee \neg R)$.

Solution.

1. When making truth tables, it's best to break these expressions up into as many component parts as possible. For this one, I'll write the truth values for P, Q, $\neg P$, $Q \vee \neg P$, $P \wedge (Q \vee \neg P)$ and finally $\neg[P \wedge (Q \vee \neg P)]$. (This might be slight overkill, but it's better to err on the side of too much detail in this class.)

P	Q	$\neg P$	$Q \vee \neg P$	$P \wedge (Q \vee \neg P)$	$\neg[P \wedge (Q \vee \neg P)]$
T	T	F	T	T	F
T	F	F	F	F	T
F	T	T	T	F	T
F	F	T	T	F	T

[1]Technically, I need parentheses here. However, you can check that $(A \vee B) \vee C \equiv A \vee (B \vee C)$, so in a long chain of "or"s, parentheses do not matter.

2. Here's another truth table, as many columns as I can stand.

P	Q	$\neg P$	$\neg Q$	$P \wedge Q$	$\neg P \wedge \neg Q$	$(P \wedge Q) \vee (\neg P \wedge \neg Q)$
T	T	F	F	T	F	T
T	F	F	T	F	F	F
F	T	T	F	F	F	F
F	F	T	T	F	T	T

3. This one's got a few more rows. (Recall that if there are n statements P, Q, etc., when we'll have 2^n rows in our truth table.)

P	Q	R	$\neg R$	$P \vee Q$	$P \vee \neg R$	$(P \vee Q) \wedge (P \vee \neg R)$
T	T	T	F	T	T	T
T	T	F	T	T	T	T
T	F	T	F	T	T	T
T	F	F	T	T	T	T
F	T	T	F	T	F	F
F	T	F	T	T	T	T
F	F	T	F	F	F	F
F	F	F	T	F	T	F

□

Problem 7. Find a formula using only $\wedge$ and $\neg$ that is equivalent to $P \vee Q$. Justify your answer with a truth table.

Solution. Instead of repeating the truth table here, I'm going to make use of Problem 9.1, which is one of De Morgan's laws. Notice that

$$P \vee Q \equiv \neg\neg(P \vee Q) \equiv \neg(\neg P \wedge \neg Q),$$

with the first equivalence by the double negation law and the second by De Morgan's law. Therefore $P \vee Q \equiv \neg(\neg P \wedge \neg Q)$. □

Problem 8. Find a formula using the connectives $\wedge$, $\vee$, and $\neg$ that has the following truth table.

P	Q	???
T	T	F
T	F	T
F	T	T
F	F	F

Solution. After some exploration, I claim that a correct solution is $(P \vee Q) \wedge \neg(P \wedge Q)$. (This is "exclusive or," which means "P or Q, but not both P and Q.") Here's a truth table.

P	Q	$P \vee Q$	$P \wedge Q$	$\neg(P \wedge Q)$	$(P \vee Q) \wedge \neg(P \wedge Q)$
T	T	T	T	F	F
T	F	T	F	T	T
F	T	T	F	T	T
F	F	F	F	T	F

Therefore this is the truth table for $(P \vee Q) \wedge \neg(P \wedge Q)$. (You could come up with an equivalent expression.)

There are of course infinitely many correct answers. Another possibility is $(P \wedge \neg Q) \vee (\neg P \wedge Q)$. □

Problem 9. Use truth tables to prove the following.

1. $\neg(P \vee Q) \equiv \neg P \wedge \neg Q$. (One of DeMorgan's laws)
2. $P \wedge (Q \vee R) \equiv (P \wedge Q) \vee (P \wedge R)$. (One of the distributive laws)

Solution.

1. The following truth table shows the desired equivalence.

P	Q	$\neg P$	$\neg Q$	$P \vee Q$	$\neg(P \vee Q)$	$\neg P \wedge \neg Q$
T	T	F	F	T	F	F
T	F	F	T	T	F	F
F	T	T	F	T	F	F
F	F	T	T	F	T	T

   Since $\neg(P \vee Q)$ and $\neg P \wedge \neg Q$ always have the same truth values, this shows that $\neg(P \vee Q) \equiv \neg P \wedge \neg Q$.

2. The following truth table shows the desired equivalence.

P	Q	R	$Q \vee R$	$P \wedge Q$	$P \wedge R$	$P \wedge (Q \vee R)$	$(P \wedge Q) \vee (P \wedge R)$
T	T	T	T	T	T	T	T
T	T	F	T	T	F	T	T
T	F	T	T	F	T	T	T
T	F	F	F	F	F	F	F
F	T	T	T	F	F	F	F
F	T	F	T	F	F	F	F
F	F	T	T	F	F	F	F
F	F	F	F	F	F	F	F

   Since $P \wedge (Q \vee R)$ and $(P \wedge Q) \vee (P \wedge R)$ always have the same truth values, this shows that $P \wedge (Q \vee R) \equiv (P \wedge Q) \vee (P \wedge R)$.

□

Problem 10. For each of the following, use truth tables to determine whether the expression is a tautology, a contradiction, or neither.

1. $\neg Q \vee (P \to Q)$.
2. $(Q \wedge P) \wedge (P \to \neg Q)$.
3. $Q \vee (P \wedge \neg Q)$.
4. $\neg Q \to (P \wedge \neg P)$.

Solution.

1. Consider the following truth table.

P	Q	$\neg Q$	$P \to Q$	$\neg Q \vee (P \to Q)$
T	T	F	T	T
T	F	T	F	T
F	T	F	T	T
F	F	T	T	T

   Since all possible truth values are true, the expression $\neg Q \vee (P \to Q)$ is a tautology.

2. Consider the following truth table.

P	Q	$\neg Q$	$Q \wedge P$	$P \to \neg Q$	$(Q \wedge P) \wedge (P \to \neg Q)$
T	T	F	T	F	F
T	F	T	F	T	F
F	T	F	F	T	F
F	F	T	F	T	F

Since all possible truth values are false, the expression $(Q \wedge P) \wedge (P \to \neg Q)$ is a contradiction.

3. Consider the following truth table.

P	Q	$\neg Q$	$P \wedge \neg Q$	$Q \vee (P \wedge \neg Q)$
T	T	F	F	T
T	F	T	T	T
F	T	F	F	T
F	F	T	F	F

Since the truth values are neither all true nor all false, the expression $Q \vee (P \wedge \neg Q)$ is a neither a tautology nor a contradiction.

4. Consider the following truth table.

P	Q	$\neg P$	$\neg Q$	$P \wedge \neg P$	$\neg Q \to (P \wedge \neg P)$
T	T	F	F	F	T
T	F	F	T	F	F
F	T	T	F	F	T
F	F	T	T	F	F

Since the truth values are neither all true nor all false, the expression $\neg Q \to (P \wedge \neg P)$ is a neither a tautology nor a contradiction.

□

Problem 11. Let P be the statement "He is van Gogh" and Q be the statement "He cut his ear." Write the conditional statement $P \to Q$ in English using the following terms.

1. Using the words "if" and "then."
2. Using the word "implies."
3. Using the phrase "only if."
4. Using the phrase "is necessary for."
5. Using the phrase "is sufficient for."

Solution.

1. If he is van Gogh, then he cut his ear.
2. Him being van Gogh implies that he cut his ear.
3. He is van Gogh only if he cut his ear. (I.e., he cannot be van Gogh without having cut his ear.)
4. Him having cut his ear is necessary for him to be van Gogh. (Same idea as above.)
5. Him being van Gogh is sufficient for him to have cut his ear.

□

Problem 12 (Practice.)**.** For extra practice, I recommend completing the progress checks in 1.1, 2.1 and 2.2. Do not hand these in. Just think about the problems on your own. You can look up solutions in the back of the book.